Coordinate methods and triangle congruence

ANSWER KEY

Core Concepts: Coordinate methods and triangle congruence

1 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 101° ✓

b. 111°

c. 93°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 45^\circ) = 101^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 9

c. 7 ✓

d. 10

Substituting $x = 1$ into $y = 3x + 4$ gives $y = 3(1) + 4 = 7$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 52^\circ) = 94^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm ✓

c. 32 cm

d. 72 cm

Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 7

b. 12

c. 10 ✓

d. 13

Substituting $x = 2$ into $y = 3x + 4$ gives $y = 3(2) + 4 = 10$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 59^\circ) = 87^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 10

b. 15

c. 13 ✓

d. 16

Substituting $x = 3$ into $y = 3x + 4$ gives $y = 3(3) + 4 = 13$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying coordinate methods and triangle congruence, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 80° ✓

b. 90°

c. 72°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 66^\circ) = 80^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm ✓

c. 48 cm

d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 13

b. 18

c. 16 ✓

d. 19

Substituting $x = 4$ into $y = 3x + 4$ gives $y = 3(4) + 4 = 16$.

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